PROJECT AND TEAM INFORMATION

Project Title

PulsePath : Smart Routing for Faster Emergency Ambulance Response

Student / Team Information

Team Name:	***Code Prodigies***
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words)

Emergency situations require an ambulance to reach the patient as quickly as possible. Choosing an ambulance only by distance does not always give the fastest response. The nearest ambulance may already be busy, or the road may have traffic or be closed.
We chose this project to develop a system that can select a suitable available ambulance and find a fast route to the emergency location. The city road network will be represented as a graph, where locations are nodes and roads are weighted edges. The weight of an edge will represent the estimated travel time for that road.
The system will check the available ambulances and calculate which one can reach the emergency location in the least time. After reaching the patient, it will also find a suitable route to a nearby hospital.
Another important part of the project is handling changes in road conditions. If traffic increases or a road is closed, the system can update the road information and calculate another route. Through this project, we want to apply graph data structures and shortest-path algorithms to a practical emergency routing problem.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Ambulance services currently use GPS, communication centres and routing systems to manage ambulances and guide them to emergency locations. In simpler systems, an ambulance may be assigned manually or the nearest available ambulance may be selected. However, the nearest ambulance is not always the fastest option.
Traffic, accidents and road closures can also increase travel time and make a selected route less suitable. Routing systems can use road networks represented as graphs and shortest-path algorithms to find better routes.
Our project follows this graph-based approach but focuses on ambulance selection and emergency routing. It considers both ambulance availability and travel time instead of only distance. Dijkstra's algorithm will be used to find minimum-time routes. When traffic conditions or road closures change, the road weights can be updated and the route can be calculated again.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The primary goal of this project is to develop a smart ambulance dispatch and routing system that helps reduce the time required to respond to an emergency. The system will model the city road network as a weighted graph and use efficient graph algorithms to make routing decisions. The main goals of the project are:

- *To identify and select the optimal available ambulance for an emergency.*
- *To calculate the minimum travel-time route using Dijkstra's algorithm.*
- *To find the nearest hospital after the ambulance reaches the patient.*
- *To handle changes such as traffic congestion and road closures through dynamic route updates.*
- *To use efficient data structures such as adjacency lists and priority queues for better performance.*
- *To reconstruct and display the actual path between locations.*

The project will be completed through several milestones. The first stage involves designing the city road network using graph representation. Next, ambulance and hospital information will be integrated into the system. The shortest-path engine based on Dijkstra's algorithm will then be implemented. After this, ambulance selection and hospital selection modules will be developed. A dynamic traffic handling feature will be added to update road conditions. Finally, the complete system will be tested using different emergency scenarios, including unavailable ambulances and closed roads.

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

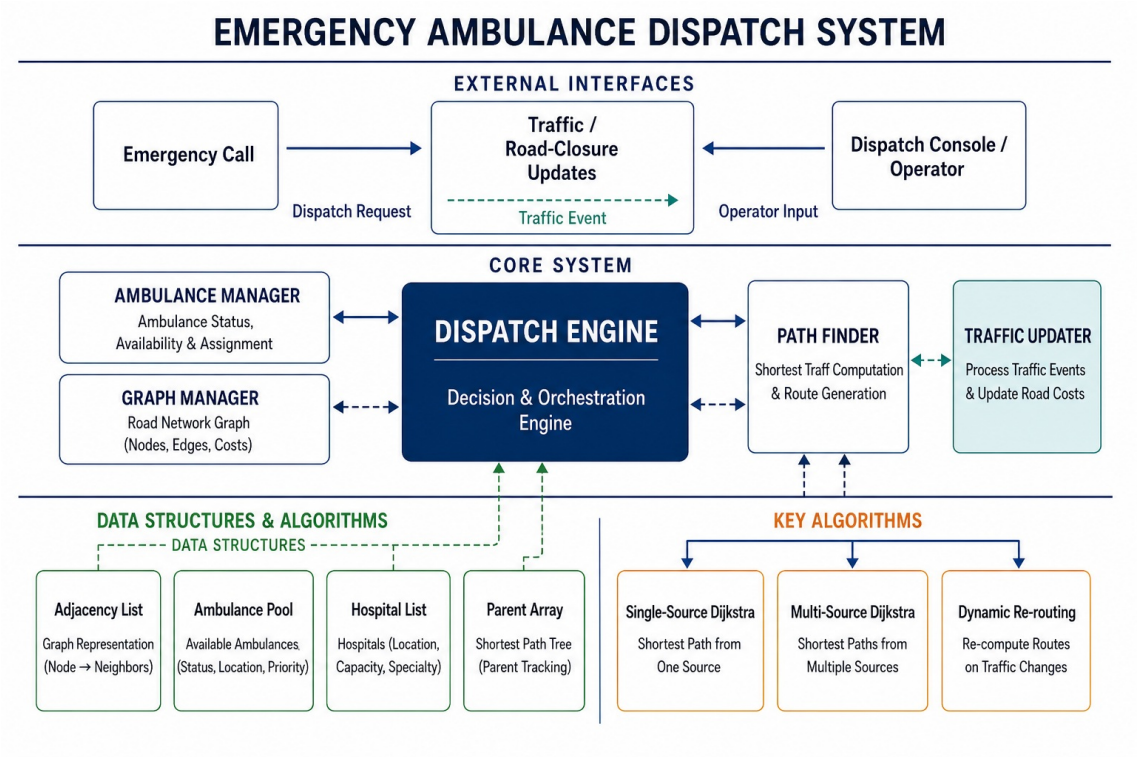
The system will be developed in C++ using graphs and shortest-path algorithms. The city road network will be stored as a weighted graph. Each location will be represented by a node and each road by an edge. The weight of each edge will represent the estimated travel time.

We will use an adjacency list to store the graph because it is suitable for a road network where most locations are connected to only some of the other locations. A priority queue will be used with Dijkstra's algorithm to select the location with the lowest current travel time. When an emergency request is received, the system will first check which ambulances are free. It will then calculate the travel time from each available ambulance to the emergency location and select the ambulance with the minimum travel time.

Once the ambulance reaches the patient, the system will calculate routes to the available hospitals and select the one with the minimum travel time. A parent array will be used to reconstruct the actual route. If traffic changes or a road is closed, the corresponding road weight will be updated. The system will then run the shortest-path calculation again to find a better route.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The project is expected to provide a practical solution for improving ambulance dispatch and emergency routing using graph algorithms. The system will focus on selecting an available ambulance, finding the fastest route, and adapting to changing road conditions. The main deliverables of the project are:

- *A functional C++ based prototype for smart ambulance dispatch and routing.*
- *A weighted graph representation of the city road network.*
- *A system to track ambulance locations and availability status.*
- *Automatic selection of the most suitable free ambulance.*
- *Fastest route calculation using Dijkstra's shortest-path algorithm.*
- *Identification of the nearest hospital based on travel time.*
- *Dynamic route recalculation for traffic changes and road closures.*
- *Efficient implementation using adjacency lists and priority queues.*

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

The system is developed based on the following assumptions:

- *All road travel times are non-negative.*
- *Ambulance locations and availability status are updated correctly.*
- *Hospital locations are predefined in the system.*
- *The city road network is accurately represented as a weighted graph.*
- *Traffic changes and road closures are provided as updates to road weights.*
- *The current system handles one emergency request at a time.*
- *A valid route exists between the required locations.*

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. *E. W. Dijkstra, "A Note on Two Problems in Connexion with Graphs," Numerische Mathematik, vol. 1, pp. 269–271, 1959.*

2. *T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, Introduction to Algorithms, MIT Press.*

3. *B. Stroustrup, The C++ Programming Language, Addison-Wesley.*

4. *GeeksforGeeks – Dijkstra's Shortest Path Algorithm using Priority Queue. available on : https://www.geeksforgeeks.org/dsa/dijkstras-shortest-path-algorithm-using-priority$_q$ueue − stl/*

5. *OpenStreetMap.Available : https://www.openstreetmap.org*